

Electric Ride Booking Recommendation System

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Abstract

Smart Cities (or Cities 2.0) are development in citizen habitation . Nowadays usage of private vehicles per single passenger not looks convenable in sustainable Smart Cities (SC) because of vehicles' allocation and pollution. As India comes under top 10 global emitters of green house gases . To regulate the effect of pollution government started promoting Electric vehicles for ride hailing . Ev's play a crucial role in achieving sustainable development goals(SDG). Based on mobility patterns, proposed a solution from which we can dynamically predict routes along with it fare prices. This system will assists user to choose a nearby driver in best possible minimum wage .The primary objective is to develop a prediction model that determines the best minimum fare price along with shortest path .The developed application had found Random Forest is giving the best accuracy for minimum fare prices and giving optimized results.

Keywords

Smart Cities , Prediction , Ride Sharing , Recommendation , Random Forest , KNN , Artificial Intelligence , Map Integration , real time API integration

1 Introduction

Ride hailing industry is growing rapidly in today's world. In major big cities or metropolitan cities are facing severe traffic congestion due to high population density and people owning their own private vehicles because inadequate public transport facility . This increased fuel consumptions and stress to commuters. As inconsistent fare pricing and not getting comfortable rides in a peak hour often frustrate commuters . This sector's growth is fueled by factors such as increased consumer spending power and a focus on convenience, affordability, and comfort, making it a critical player in the urban transportation landscape[1]. By narrowing the research scope to ride-hailing, this research project aims to provide a practical and immediate solution to urban transportation challenges, empowering riders to conveniently compare and select the most suitable ridesharing options [2].

As a result, cities would be able to better support people's travel demand through a regulated, efficient, more sustainable public transportation system. Meanwhile, with the wide deployment of Automatic Fare Collection (AFC) systems on bus networks and Global Positioning System (GPS) devices on taxis, large amounts of taxi traces are collected, with which we can provide integrated analysis on the mobility patterns of both public and private transportation[3].

There is not a significant amount of work focusing on ev's (Electric Vehicles) booking online application, where riders can see all ride-sharing options, prices, and offers at once and can select best convenient mode as per their requirements. Therefore this research is focused on designing a mobile application that can provide the information in a more convenient way to riders and gives them option to choose and compare the best option.

2. Related Works

In this section, first we discuss the related works in recommending fare price problem.

2.1 Fare Price Approaches

In [4], The article describes the implementation and evaluation of four different types of machine learning models (Logistic Regression, Decision Trees, K-Nearest Neighbors, and Random Forest) to estimate the probability of booking (the best rate) and found that the Random Forest model gave the best results in the two evaluation methods used.

The most notable feature of this article is that four machine learning models were developed: Logistics Regression, Decision Tree, K-Nearest Neighbor (KNN), and Random Forest Classifiers; the Random Forest Classifiers had the greatest AUC-ROC scores.

The Random Forest Classifiers had a mean AUC-ROC score of 0.773 and 0.781 when evaluated using a ten-fold three-way split and a ten-fold cross-validation, respectively. Additionally, standardization of the data improved the mean AUC-ROC scores and the standard deviation of both the Logistics Regression and KNN models. Data standardization helped improve the AUC-ROC scores of Logistics Regression and KNN. It was believed that the improvements in score and decreases in time for development of the Logistics Regression model led to the increased AUC-ROC scores. We utilized two evaluation techniques to calculate AUC-ROC scores for all models, ten-fold three-way splits and ten-fold cross-validations.

The **results** of the study show that the Random Forest Classifiers outperformed other models, with a 15.03% higher AUC-ROC score than Decision Tree, 2.93% higher than KNN, and 2.38% higher than Logistics Regression.

In[5], The Random Forest algorithm allows for routing of public transport ways through its use for optimizing routes, predicting passenger demand, and dynamically adjusting fares.

This study demonstrates that the system proposed can make improvements to the way metropolitan transportation systems operate in terms of improving customer service, reducing the amount of time a passenger has to wait, and increasing the overall efficiency of the transportation system. By leveraging machine learning techniques and cloud-based services, these types of transportation systems have the ability to become more responsive to their customers.

As noted in the Findings section of the paper: One major finding is that the passenger experience has improved greatly due to less time spent waiting and an increase in the overall efficiency of the trip. There was also a significant increase in the efficiency of daily operations with accuracy rates at around 90% and mean absolute error rates falling below 4 consistently. As demonstrated by these findings, the Random Forest algorithm will improve the efficiency of operations, decrease the amount of time a passenger spends waiting, and ultimately improve customer service.

In summary, the Random Forest algorithm provides reliable predictions; it holds steady around a 90% accuracy rate and a mean absolute error below four.

Another important feature of the Random Forest algorithm is its ability to respond in real time to forecasted events and immediately modify routes; thus, making it an integral part of any system that continuously strives to provide better service to its customers. It is evident from these studies that the Random Forest algorithm will become the fundamental technology used to develop more dynamic and creative solutions for public transportation systems, which is essential as urbanization grows.

The methods used in the study include data collection, data preprocessing, Random Forest training, decision making, optimization and adaptation, application in transit system, evaluation and refinement, and implementation in cloud infrastructure.

The results of the study show a significant improvement in the passenger experience, with reduced wait times and improved travel efficiency, as well as a notable increase in operational efficiency, with sustained accuracy rates approaching 90% and mean absolute error scores constantly below four.

In[6], The paper presents the design and implementation of a digital dashboard interface for electric vehicles using advanced simulation technologies and navigation APIs, providing critical information to the driver and enhancing safety and efficiency.

The system's adaptability to various vehicle models with minimal modifications and its potential for future enhancements, such as integrating additional sensors and incorporating AI algorithms for improved performance, are key findings of the study. ↓

The key findings of the study highlight the potential for future enhancements, such as integrating additional sensors and improving AI algorithms to further refine the system's capabilities. The integration of GMap.NET, OpenWeatherMap API, and Geoapify API has enabled the creation of a robust and interactive dashboard that offers comprehensive navigation, weather updates, and vehicle performance metrics.

This research underscores the importance of real-time data processing and user-friendly interfaces in modern electric vehicles, offering a scalable and cost-effective solution for enhancing vehicle usability and driver satisfaction

The proposed digital dashboard interface represents a significant advancement in electric vehicle technology, contributing to safer, more efficient, and enjoyable driving experiences.

The simulation program demonstrated the system's capabilities, including real-time updates of vehicle parameters such as speed, battery level, and remaining range, providing an accurate representation of the vehicle's performance.

The results of the study demonstrate a significant improvement in providing real-time vehicle data and enhancing the overall driving experience through a user-friendly interface. The simulation program provides an accurate representation of vehicle performance and assists in route planning and energy management.

3. Proposed Work

3.1 Overview of Proposed System

The suggested system is named "EvRide -Smart Electric Vehicle Ride", an artificial intelligence-enabled ride booking application exclusively for electric vehicles (EVs). As opposed to the existing ride booking applications like Uber and Ola, which primarily cater to fuel-consuming vehicles, EvRide caters only to optimizing electric rides through smart route planning, energy-saving ridesharing and intelligent fare predictions based on machine learning algorithms. The web-based system enables users to book rides of electric vehicles through a responsive web portal, ensuring effective use of energy sources as well as accurate fare estimation. The back end of the system is developed using FastAPI with the use of Random Forest Regression for fare estimation and battery usage. The front end is developed using HTML, CSS, and JavaScript.

3.2 System Architecture and Design

The system architecture of *EvRide* follows a three-tier structure comprising the following layers:

- Presentation Layer(Frontend)
- Application Layer (Backend – FastAPI)
- Data Layer (SQLite Database)

System Architecture Flow:

Following the user authentication stage in fig1, the client sends the coordinates of the starting point and the ending point to the FastAPI backend through a REST API call. The backend makes a request for the acquisition of additional information through both internal and external resources to obtain context information such as traffic density, distance of the trip, battery level, among others. All of these features will be used in the data preprocessing stage before being input to the machine learning model which will provide the results of the fare, estimated energy expenditure, and the time of the ride. Finally, all of the results will be returned from the backend through an API response to the client. After the user accepts the offer, the backend stores all the information into the database.

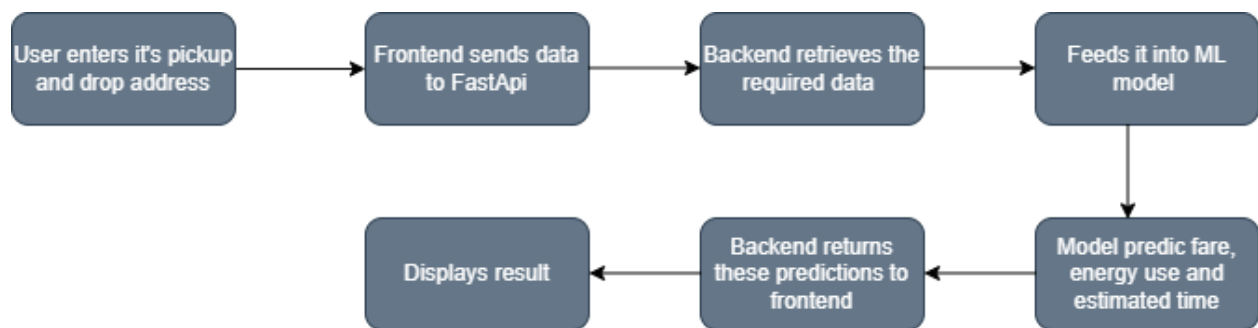


Fig 1, shows the workflow of application

3.3 DataSet Description

EVRide Fare Prediction Data Set is an electric vehicle ride data set curated for predicting dynamic pricing in rideshare services. This data set includes various practical factors that affect the change in fares of rides in different metropolitan cities of India. The data set consists of 5,000 rides that have been taken using electric vehicles with specific features such as distance, time, vehicle efficiency, demand, and context.

The data set has been constructed by considering several aspects of the real-world electric vehicle-based transportation system. Temporal factors include time of day, hour, and holidays; environmental factors include weather, temperature, and humidity; EV-specific factors include battery condition, power, charge station availability, and interaction between passenger and driver factors such as rider type, driver rating). Collectively, these features offer a well-rounded basis for predicting fare in an electric ride-sharing system.

The following eight categories of features are included in the data set:

- Trip Features – which include trip length (km), duration of the trip (min), difficulty level of the route, and traffic intensity.
- Demand & Surge Features – included in this feature set are factors related to demand, surges, and historical fares.
- EV Specific Features – among these features is EV battery life (%), kWh consumed, and proximity to charger stations.
- Temporal Features – these features include time-of-day (Morning, Afternoon, Evening, Night), day of week, and holiday status indicator.
- Environmental Features – this includes weather conditions such as sunny, foggy, rainy, hot, stormy, temperature (C), and humidity (%).
- User & Vehicle Features – this includes the type of vehicle such as Sedan, SUV, Compact, and premium versus regular customer.
- Geographical Feature – indicates the city where the rides take place Delhi, Mumbai, Pune, Bangalore, Chennai.
- Output Feature (target) – this is the fare amount collected (INR).
- In this project, only the tabular ride level features were used for predicting fare amounts without any additional location-based features and sensor-derived features.

A summary of the dataset structure is provided in *Table 1*:

Dataset	Instances	Features	Target
EvRide Fare Dataset	5000	21 input features	fare_amount_inr

The data set can effectively be used in various machine learning problems including price optimization, dynamic pricing, and demand forecasting. The wide range of characteristics provided makes it possible for models to generalize from more complicated interactions, hence making better predictions based on varying circumstances during a trip. Moreover, the inclusion of electric vehicle specific characteristics makes it relevant for electric mobility analysis.

3.4 Dataset Preprocessing

In order to ensure the reliability and quality of predictive modeling, a comprehensive dataset preprocessing pipeline was followed before the start of modeling. The EVRide dataset consisted of a total of 5,000 rides with numeric, categorical, temporal,

environmental, and EV-related characteristics. In order to prepare this unstructured data to be used for machine learning purposes, a number of preprocessing steps were undertaken. These are described below.

1. Data Cleaning: The presence of irrelevant data such as trip_id was first checked and removed as it does not contribute anything towards fare prediction.

A detailed check for missing data was conducted and there were no missing data points. As such, no further imputation of missing values was required in this case.

2. Outlier Identification and Removal: As the target variable of fare_amount_inr had an extremely wide distribution range and some outliers caused due to very long trips or surge price peaks, it was decided to perform outlier removal.

Outliers on the basis of top and bottom one percent quantiles were identified and removed.

3. Feature Type Classification: All the features were classified based on the type of their data:

- Numerical Features: distance, duration, traffic level, battery condition, energy consumed, temperature, humidity, etc.
- Categorical Features: city, weather condition, time of day, day of week, vehicle type, user type, etc.
- Binary Features: is_holiday.
- Features with High Cardinality: those that have many unique values.
- Features with Low Cardinality: those that have no more than 10 unique values.

This classification provided the grounds for choosing adequate encoding schemes at subsequent stages.

4. Categorical Feature Encoding: Since any model requires numerical input, all categorical features underwent encoding with appropriate schemes:

- One-Hot Encoding (OHE) was used to encode features with low cardinality (such as weather_condition, time_of_day, vehicle_type) into separate binary columns.
- Label Encoding was used to encode features with high cardinality, to prevent increasing the dimensionality too much (city after the filtering stage).

Both methods were combined in a ColumnTransformer to apply consistently to all models.

5. Scaling of Numerical Features: While models such as Random Forest and XGBoost do not necessarily need scaling, algorithms including KNN and SVR proved to be extremely sensitive to scaling. For this reason, numerical features were encoded with StandardScaler, which scales the features according to the z-scores.

6. Dealing With Imbalanced Classes: In Temporal And Environmental Features: The temporal and environmental features were slightly imbalanced because some categories occurred more often than others. However, One-Hot encoding helped prevent any class imbalance from having any effect on how the model performed.

7. Final Processed Feature Matrix: As a result of the preprocessing, our final dataset had:

- Numerical feature matrix
- No missing data
- Minimized outliers
- Processed features
- Final feature selection using feature importance in random forest

This well-organized dataset made it easy to carry out tasks like Random Forest, XGBoost, KNN regression, and support vector regression.

3.5 Model Selection

Of all the algorithms used in this study for machine learning models including KNN Regression, SVR Regression, XGBoost Regression, and Random Forest Regression, Random Forest is the best algorithm in terms of performance in the task of predicting EV fares. Using the right form of preprocessing and feature engineering and proper hyperparameters, Random Forest model obtained an R^2 score of 0.9165 which is very high, and means that the model is able to explain over 91% of the variance in the fare values. Moreover, Random Forest generated very small error metrics including MAE and RMSE.

The best model was the Random Forest due to its capability of dealing with non-linear data, high dimensional data, as well as numerical and categorical variables without extensive parameter adjustment. The use of many decision trees gives it an ensemble capability that enables it to model interactions between features, which include distance, traffic, surge factor, road conditions, and environmental factors. Additionally, Random Forest is less sensitive to noise and outliers since, in most cases, it is a bagging algorithm; hence, it can be used to predict prices in a dynamical system of EVs based on fare requests.

In addition, the model clearly ranked the importance of each feature, which made it more interpretable than other complicated models, such as SVR or KNN. This made it easier to determine significant predictors, including distance, time, demand ratio, battery status, surge factor, and traffic condition. It is evident that due to its high accuracy, Random Forest is suitable for implementation in the fare prediction engine at EVRide.

From the analysis, it is clear that Random Forest is the best model in predicting the fare using the given data set at EVRide.

3.6 Experimental Setup

1. Coefficient of Determination (R^2 Score): R^2 measures how much of the variance in the actual target values is explained by the model.

It indicates the goodness of fit of the regression model.

Formula:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

2. Mean Absolute Error (MAE): MAE measures the average magnitude of prediction errors, without considering their direction.

It directly expresses how many units (₹) the predictions are off on average.

Formula:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

3. Root Mean Squared Error (RMSE): RMSE measures the square root of the average squared error between actual and predicted values.

It penalizes larger errors more heavily than MAE.

Formula:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

4. Accuracy (Regression Accuracy): Since regression does not use classification accuracy, a normalized accuracy measure is used to express how close predictions are to actual values.

Formula:

$$\text{Custom Accuracy} = 1 - \frac{MAE}{\bar{y}}$$

4. Results

In this section, results from the experiments on performance evaluation of the machine learning algorithms in the context of the EVRide fare prediction data set have been illustrated. Four different regression algorithms have been considered, namely Random Forest, XGBoost, K-nearest neighbor, and Support vector regression (SVR). Evaluation has been done based on traditional parameters such as R2 score, mean absolute error, root-mean-square error, and custom accuracy, which is 1- MAE/mean(fare).

4.1 Model Performance Comparison

As can be seen from the experimental results, all four models vary greatly in terms of their forecasting performance. The model that yielded the highest performance was Random Forest Regression, which returned the highest R2 score of 0.9165. This means that the model was able to account for about 91 percent of the variance in fares. Low values of MAE and RMSE confirm that the predictions provided by the model were stable, with lower errors compared to the actual data. Random Forest Regression was particularly advantageous due to its noise tolerance.

Close behind was **XGBoost Regression**, which performed competitively and showed robust non-linear learning capabilities. While XGBoost Regression usually produces better performance compared to decision trees when applied in pricing scenarios,

Random Forest Regression produced slightly more stable results and smaller prediction errors in this particular dataset. This performance indicates that the given data was relatively small and that its features were highly understandable, therefore, suitable for the bagging ensemble approach used in Random Forest.

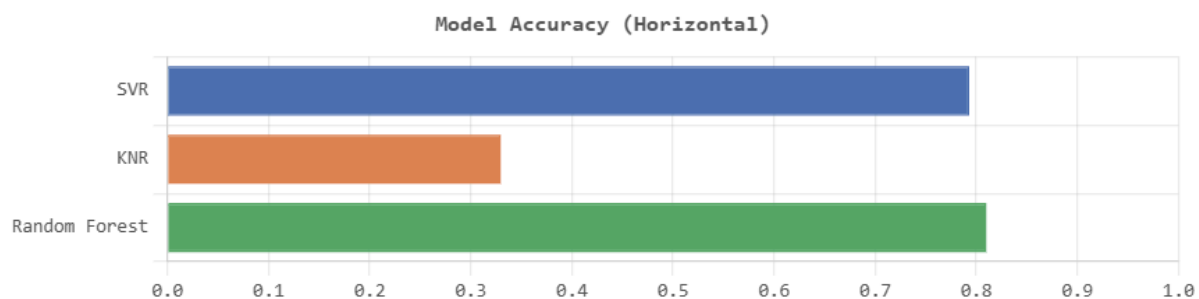
Finally, the **KNN regressor** produced mediocre results and was inferior to tree-based models. The model requires scaled and normalized values since it heavily depends on the proximity of the samples. KNN performed poorly in recognizing complex non-linear relationships related to surge pricing, battery impact, and weather conditions on the number of rides.

Among all the other models, **SVR model** was the least efficient predictor model. Even though SVR works efficiently on small data sets and linear data sets, in the case of multiple non-linear interaction data sets, it gets more complicated. Hence, SVR model performs relatively poorly in this case.

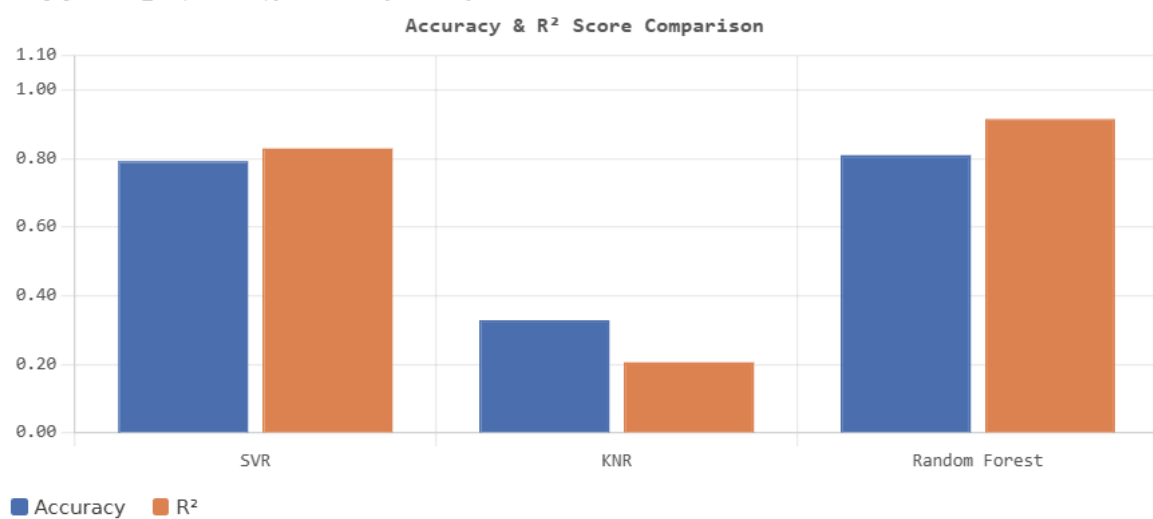
Table 4.1 Comapritive results of various machine learning models

Model	Accuracy	R^2	RMSE	MAE	Remarks
SVR	0.7931	0.83	747.37	368.29	Simple and Interpretable
KNR	0.3297	0.2061	1640.48	1193.46	Performs poorly due to high sensitivity
Random Forest	0.81	0.9165	531.93	328.77	Highest accuracy robust and reliable

In [3]: model_comparison.py – Overall Accuracy (Horizontal)



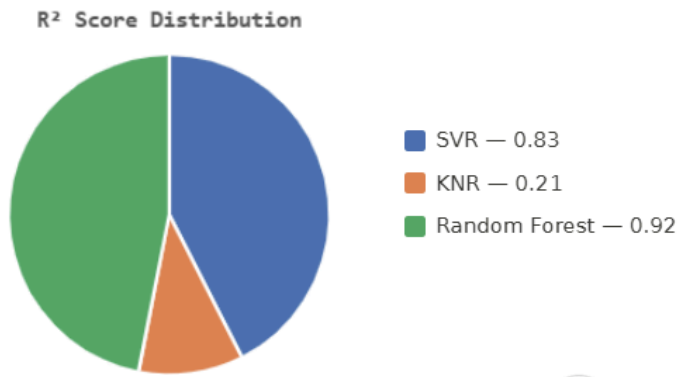
In [1]: model_comparison.py – Accuracy & R^2 by Model



In [2]: model_comparison.py – RMSE & MAE by Model



In [4]: model_comparison.py – R² Score (Pie)



5. Conclusion

In this study, an intelligent fare prediction system for electric vehicle ride-hailing, EVRide, was developed using a range of machine learning regression techniques. The goal was to make accurate ride fare estimates based on an amalgamation of trip attributes, environmental factors, demand trends, and EV-specific operational parameters. Experimental results demonstrate that **Random Forest Regression** outperformed all other models with the highest R² score of 0.9165 as well as the lowest MAE and RMSE values. This suggests that the model was able to effectively learn the underlying patterns of fare variation with high precision and consistency. The ensemble nature of Random Forest enabled it to successfully model complex non-linear relationships while being robust against noise and outliers present in real-world ride data.

XGBoost Regression had a competitive performance and a strong capability in dealing with non-linear interactions but Random Forest gave more consistent predictions with slightly lower error rates in this data set. On the other hand, **KNN Regressor** and **SVR** showed relatively less performance. KNN is sensitive to feature scaling and high-dimensional data, while SVR is not very effective in capturing the complex interaction among multiple features.

Our results clearly show that ensemble-based tree models are very suitable for dynamic pricing problems in electric ride-hailing systems. We combined different features such as traffic conditions, surge multiplier, battery health and environmental features that helped the model to learn real-life pricing scenarios.

In summary, the EVRide system exemplifies the promise of machine learning to improve fare prediction accuracy, increase transparency, and support data-driven decision

making in the domain of electric mobility platforms. The proposed approach can be extended and embedded into real-world ride-hailing applications to provide more reliable and optimized pricing strategies.

This work demonstrates the potential of machine learning to revolutionize the pricing mechanisms in next generation systems of electric mobility.

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